

Project Report

This project was completed as part of the CodeAlpha Data Analytics Internship to understand the fundamentals of web scraping using Python. The main objective was to collect data from a public website and organize it into a structured dataset for further analysis.

For this project, I scraped quotes, author names, and related tags from the **Quotes to Scrape** website. After extracting the data, I organized it into a clean dataset that can be used for Exploratory Data Analysis (EDA) and data visualization.

Data Source

Website: <https://quotes.toscrape.com/>

This website is specially designed for learning and practicing web scraping techniques, making it an ideal source for this project.

Tools and Technologies Used

- Python
- Requests
- BeautifulSoup
- Pandas
- Google Colab

Project Workflow

I started by importing the required Python libraries, including **Requests** for sending HTTP requests, **BeautifulSoup** for parsing HTML content, and **Pandas** for organizing the extracted data.

Next, I scraped data from all 10 pages of the website. For each quote, I extracted the quote text, the author's name, and the associated tags. The collected information was then stored in a Pandas DataFrame with three columns: **Quote**, **Author**, and **Tags**.

After organizing the data, I checked the dataset structure using `df.shape` and `df.info()` to ensure the records were complete and there were no missing values. Finally, I exported the dataset as a CSV file named **Quotes_Data_100.csv** for future analysis.

Dataset Information

- **Total Records:** 100
- **Total Columns:** 3

The dataset includes the following columns:

- Quote
- Author
- Tags

All 100 records were successfully extracted, and the dataset contains no missing values.

Output

The final output of this project is a CSV file named **Quotes_Data_100.csv**, which contains the scraped data in a structured format. This dataset can be used for further tasks such as Exploratory Data Analysis (EDA), data visualization, and text analysis.

Conclusion

Through this project, I gained practical experience in web scraping using Python. I learned how to collect data from a website, extract useful information, organize it into a structured dataset, and save it for future analysis. This project strengthened my understanding of web scraping, HTML parsing, and basic data handling techniques, which are essential skills in data analytics.